1681: Mineral salt medium with colloidal chitin

Final pH: 7
Final volume: 1000 ml

Solution A	50.00	ml
Solution B	50.00	ml
Modified Wolin's mineral solution	10.00	ml
Wolin's vitamin solution	10.00	ml
Sodium resazurin (0.1% w/v)	1.00	ml
Yeast extract	1.00	g
Polypeptone	10.00	g
$NaHCO_3$	3.52	g
Colloidal chitin	5.00	g
Distilled water	890.00	ml

1. **Reducing agents**: Flush each solution under N_2 for 30 minutes, stopper and autoclave. L-Cysteine-HCL x H_2O solution (10%): add 12.5 µl per 10 ml medium Na_2S x 9 H_2O solution (3%): add 40 µl per 10 ml medium.

2. **Colloidal chitin preparation**: Cut 5 gram commercial crab (shrimp) shell flakes into small pieces, and dissolve in 100 ml of 12M HCL while stirring in a fume hood at room temperature. Add chitin-HCL mixture slowly into pre-cooled water to obtain the colloidal chitin, centrifuge and wash in pre-cooled water till the colloidal chitin reaches a pH of 6 to 6.5. Mix 0.5% with MS medium as the sole carbon source. As an alternative commercial available chitin powder can be used.

3. **Prepare MS medium with chitin**: Dissolve the ingredients of MS medium (except $NaHCO_3$ and vitamins), boil medium under CO_2 until the resazurin indicator changes from pink to colourless. Cool down under CO_2, add $NaHCO_3$ and adjust pH to 7.0. Switch to N_2 to prevent further lowering of the pH. Fill up into Hungate tubes under N_2 and autoclave. Before use add vitamins (sterilized by filtration) and reducing agents.

Solution A

KH_2PO_4	8.00	g
NH_4Cl	5.00	g
$MgCl_2$ x 6 H_2O	4.00	g
$CaCl_2$ x 2 H_2O	1.00	g
Distilled water	1000.00	ml

Solution B

K_2HPO_4	8.00	g
Distilled water	1000.00	ml

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Wolin's vitamin solution (from medium 141)

Biotin	2.00	mg
Folic acid	2.00	mg
Pyridoxine hydrochloride	10.00	mg
Thiamine HCl	5.00	mg
Riboflavin	5.00	mg
Nicotinic acid	5.00	mg
Calcium D-(+)-pantothenate	5.00	mg
Vitamin B_{12}	0.10	mg
p-Aminobenzoic acid	5.00	mg
(DL)-alpha-Lipoic acid	5.00	mg
Distilled water	1000.00	ml

Filtrate (pore size 0.22 µm to sterilize).

Modified Wolin's mineral solution (from medium 141)

Nitrilotriacetic acid	1.50	g
$MgSO_4 \times 7\ H_2O$	3.00	g
$MnSO_4 \times H_2O$	0.50	g
NaCl	1.00	g
$FeSO_4 \times 7\ H_2O$	0.10	g
$CoSO_4 \times 7\ H_2O$	0.18	g
$CaCl_2 \times 2\ H_2O$	0.10	g
$ZnSO_4 \times 7\ H_2O$	0.18	g
$CuSO_4 \times 5\ H_2O$	0.01	g
$AlK(SO_4)_2 \times 12\ H_2O$	0.02	g
H_3BO_3	0.01	g
$Na_2MoO_4 \times 2\ H_2O$	0.01	g
$NiCl_2 \times 6\ H_2O$	0.03	g
$Na_2SeO_3 \times 5\ H_2O$	0.30	mg
$Na_2WO_4 \times 2\ H_2O$	0.40	mg
Distilled water	1000.00	ml

First dissolve nitrilotriacetic acid and adjust pH to 6.5 with KOH, then add minerals. Adjust final to pH 7.0 with KOH.

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